CSE252 · INTRODUCTION TO ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

UNIT 5

Model Evaluation and Improvement

Is your model actually any good? This unit is where accuracy stops being the answer and starts being the question.

Regression Metrics • Confusion Matrix • Precision · Recall · F1 • ROC-AUC • Cross Validation & Hyperparameter Tuning • Grid & Random Search • Explainable AI

School of Computer Science & Engineering

Session 2026-27 · CO5 · Metrics · Tuning · Explainability

OPENING STORY

The Detector That Was 99.9% Accurate and Caught Nothing

UNIT 5

The brief
flag fraudulent transactions

→

The model
one line: 'return not fraud'

→

The score
99.9% accurate

→

The catch
it detected zero fraud

HOW THE STORY GOES

A team is asked to build a fraud detector. Fraud is about 0.1% of transactions.

Here is a model `return NOT_FRAUD`. One line. It ignores its input entirely.

Its accuracy 99.9%. That is not a trick - it is arithmetic.

Frauds detected zero. Not few. Zero.

So the number was true and completely useless. Accuracy on a rare event tells you almost nothing.

DID YOU KNOW?

This is why medical screening, fraud detection and fault detection are never evaluated on accuracy. The base rate makes it meaningless.

THINK OF IT LIKE **a smoke alarm that never goes off.** right almost every night - and useless on the one that matters.

TAKEAWAY **A metric can be perfectly correct and still tell you nothing worth knowing.**

What This Unit Covers

Choosing the wrong metric is how good models get deployed badly.

- Regression metrics
  - MAE, MSE, RMSE and R^2 - and when each one lies to you
- The confusion matrix and classification metrics
  - Accuracy, Precision, Recall, F1-score, ROC-AUC
- Model improvement
  - Cross validation, hyperparameter tuning, Grid Search, Random Search
- Introduction to Explainable AI
  - Why black boxes are a problem; LIME and SHAP
- The single most important idea in the unit
  - The right metric depends on the COST of each kind of error
  - That cost is a business and ethical decision, not a technical one

What You Already Bring to This Unit

Four bridges into this unit - including one from the course you are taking alongside this one.

FROM UNIT III

Models to evaluate

- You can now build five classifiers
- This unit asks whether any of them is good
- And what 'good' even means

FROM UNIT II

Held-out data

- Train / validation / test, and why they differ
- Cross-validation and its standard deviation
- All of it becomes load-bearing today

FROM SCHOOL MATHS

Fractions and ratios

- Every metric here is a ratio of four counts
- Precision, recall and F1 need no calculus
- If you can read a fraction you can do this

WHERE IT LANDS

When errors move mass

- A missed pedestrian and a phantom stop both injure
- The threshold becomes a safety decision, not a default
- Four modules at 95% give you 81% end to end

THE FOUR BRIDGES

1. Unit III: the models
2. Unit II: honest held-out data
3. maths: four counts, some ratios
4. and on a robot, errors move mass

LEARNING OUTCOMES

Learning Outcomes

UNIT 5

By the end of this unit you should be able to:

1. **Compute** MAE, MSE, RMSE and R^2, and say when each is the right one. CO5
2. **Derive** accuracy, precision, recall and F1 from a confusion matrix. CO5
3. **Select** a metric and a threshold from the cost of each kind of error. CO5
4. **Interpret** an ROC curve and AUC, and compare two models with them. CO5
5. **Explain** a model's prediction using LIME or SHAP, and say why that matters. CO5

Remember
recall the terms

Understand
explain them in your own words

Apply
use them on a problem you have not seen

Analyse
choose between them and justify it

the levels this unit is assessed at

01

SECTION

Regression Metrics

IN THIS SECTION

- MAE - mean absolute error
- MSE and RMSE - squared error and its interpretable cousin
- R^2 - the coefficient of determination
- Adjusted R^2 - and why plain R^2 can be gamed
- Choosing a metric when outliers are present

The Four Regression Metrics

Same predictions, four different verdicts. The difference is how each metric treats a large error.

MEAN ABSOLUTE ERROR - AVERAGE SIZE OF THE ERROR, IN THE TARGET'S OWN UNITS

$$\mathrm{MAE} = (1/n) \cdot \Sigma \, | \, y_i - \hat{y}_i \, |$$

Robust to outliers. Every error counts in proportion to its size.

MEAN SQUARED ERROR AND ROOT MEAN SQUARED ERROR

$$\mathrm{MSE} = (1/n) \cdot \Sigma \, (\, y_i - \hat{y}_i \,)^2 \qquad \mathrm{RMSE} = \sqrt{\mathrm{MSE}}$$

MSE punishes large errors quadratically. RMSE returns the value to the target's units.

COEFFICIENT OF DETERMINATION

$$R^2 = 1 - (\, \mathrm{SS_res} \, / \, \mathrm{SS_tot} \,) = 1 - \Sigma (y_i - \hat{y}_i)^2 \, / \, \Sigma (y_i - \bar{y})^2$$

The proportion of variance in y explained by the model. 1.0 is perfect; 0 means no better than predicting the mean.

Worked Example: Compute All Four Metrics

THE PROBLEM

- A model predicts house prices (₹ lakh) for 5 houses:
- Actual y = 50, 60, 70, 80, 90
- Predicted $\hat{y}$ = 52, 58, 75, 78, 95
- Compute MAE, MSE, RMSE and R^2

WHY IT MATTERS

- RMSE (3.52) exceeds MAE (3.2) - it always does, because squaring weights the larger errors more
- R^2 = 0.938 means the model explains ~94% of the variance in price
- SS_res = 62 is the same sum used in MSE - only the divisor differs

STEP-BY-STEP SOLUTION

Step 1 - errors (y − ŷ) −2, +2, −5, +2, −5

Step 2 - absolute errors 2, 2, 5, 2, 5 → sum = 16

Step 3 - MAE 16 / 5 = 3.2 lakh

Step 4 - squared errors 4, 4, 25, 4, 25 → sum = 62

Step 5 - MSE and RMSE $\text{MSE} = 62/5 = 12.4 \quad \text{RMSE} = \sqrt{12.4} \approx 3.52$ lakh

Step 6 - SS_tot $\bar{y} = 70$, so $(y-\bar{y})^2 = 400, 100, 0, 100, 400 \rightarrow \text{sum} = 1000$

Step 7 - R^2 1 − (62 / 1000) = 1 − 0.062 = 0.938

ANSWER $\text{MAE} = 3.2 \cdot \text{MSE} = 12.4 \cdot \text{RMSE} \approx 3.52 \cdot R^2 = 0.938$

Adjusted R^2 - Why Plain R^2 Can Be Gamed

EXAM TIP · R^2 never decreases when you add a feature, even a useless one

The flaw add a column of random numbers to your regression and R^2 will rise - or at worst stay the same. It can never fall.

Why the model has more freedom to fit the training data, and R^2 only measures fit. It has no notion of whether the extra freedom was earned.

The consequence R^2 alone will always favour the most complex model. Optimise it and you optimise for overfitting.

Adjusted R^2 $R^2_adj = 1 - [\,(1 - R^2)(n - 1) / (n - p - 1)\,]$

How it fixes things it penalises the number of predictors p. Adding a useless feature raises p without raising R^2 enough to compensate, so Adjusted R^2 FALLS.

Use it when comparing models with different numbers of features. For a single model on a fixed feature set, plain R^2 is fine.

PREDICT I add 50 columns of pure random noise to my regression. What happens to R^2? What happens to Adjusted R^2?

02

SECTION

Confusion Matrix and Classification Metrics

IN THIS SECTION

- The confusion matrix - TP, TN, FP, FN
- Type I and Type II errors
- Accuracy, Precision, Recall and F1-score
- The accuracy paradox on imbalanced data
- ROC curves and AUC
- Choosing a metric from the cost of each error

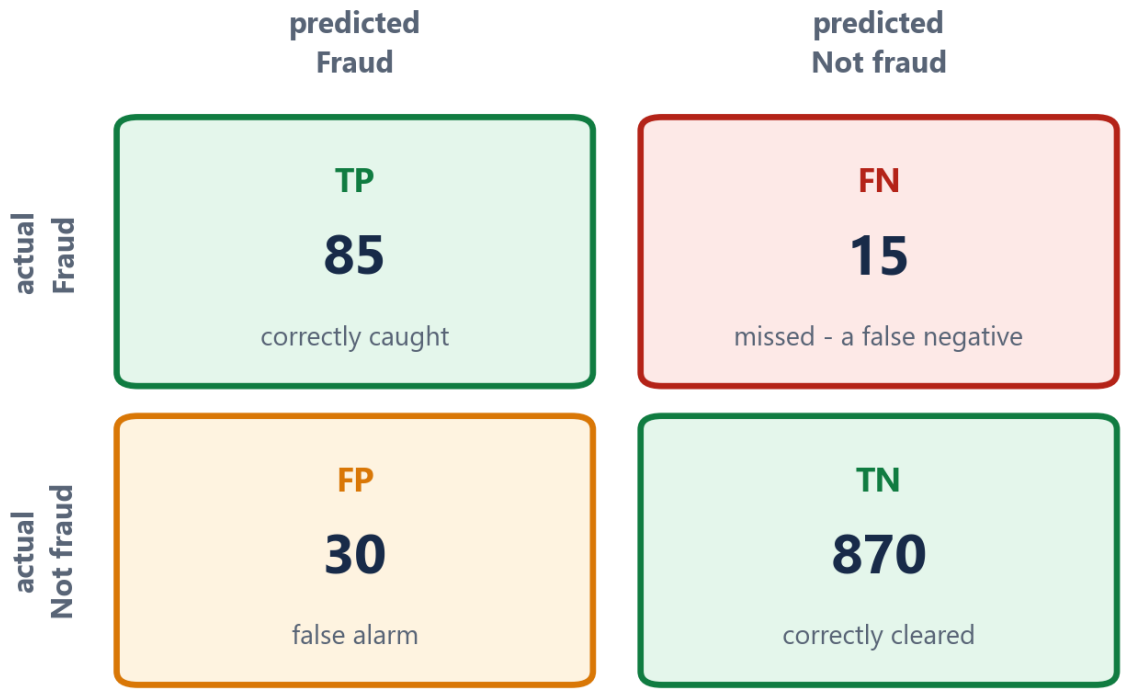
The Confusion Matrix

Four cells. Every classification metric in this unit is built from them.

	Predicted POSITIVE (1)	Predicted NEGATIVE (0)
Actual POSITIVE (1)	TRUE POSITIVE (TP) Correctly caught the positive case	FALSE NEGATIVE (FN) Missed it — Type II error
Actual NEGATIVE (0)	FALSE POSITIVE (FP) False alarm — Type I error	TRUE NEGATIVE (TN) Correctly ignored

MEMORISE IT The first word says whether the model was RIGHT. The second says what the model PREDICTED. 'False Negative' = wrongly predicted negative.

The Confusion Matrix - Where Every Metric Comes From



HOW TO READ THIS FIGURE

TP predicted fraud, and it was fraud. Correctly caught.

TN predicted not-fraud, and it was not. Correctly cleared.

FP predicted fraud, but it was legitimate. A FALSE ALARM - you blocked a real customer.

FN predicted not-fraud, but it was fraud. A MISS - the money is gone.

Everything else precision, recall, F1, accuracy - all four are arithmetic on these four cells.

TAKEAWAY **Accuracy here is 95.5% while 15 frauds went through. One number hides the whole story.**

The Four Classification Metrics

Precision and Recall answer different questions - and improving one usually worsens the other.

ACCURACY - WHAT FRACTION OF ALL PREDICTIONS WERE CORRECT

$$\text{Accuracy} = (TP + TN) / (TP + TN + FP + FN)$$

Misleading on imbalanced data. Never report it alone.

PRECISION - OF EVERYTHING FLAGGED POSITIVE, HOW MUCH REALLY WAS

$$\text{Precision} = TP / (TP + FP)$$

'How much do I trust a positive prediction?' Punishes false alarms.

RECALL (SENSITIVITY, TRUE POSITIVE RATE) - OF ALL REAL POSITIVES, HOW MANY DID WE CATCH

$$\text{Recall} = TP / (TP + FN)$$

'How much did I miss?' Punishes misses.

F1-SCORE - THE HARMONIC MEAN OF PRECISION AND RECALL

$$F1 = 2 \cdot (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

One number balancing both. Harmonic, so it is dragged down by the weaker of the two.

Worked Example: Compute Every Metric from a Confusion Matrix

THE PROBLEM

- A fraud detector is tested on 1,000 transactions.
- TP = 30 (fraud correctly caught)
- FN = 20 (fraud missed)
- FP = 50 (legitimate wrongly flagged)
- TN = 900 (legitimate correctly passed)

WHY IT MATTERS

- The dummy model BEATS the real model on accuracy while being useless
- This is the ACCURACY PARADOX - the single most important idea in this unit
- On imbalanced data, always report precision, recall, F1 and ROC-AUC

STEP-BY-STEP SOLUTION

Accuracy $(30 + 900) / 1000 = 930/1000 = 0.93 \rightarrow 93\%$

Precision $30 / (30 + 50) = 30/80 = 0.375 \rightarrow 37.5\%$

Recall $30 / (30 + 20) = 30/50 = 0.60 \rightarrow 60\%$

F1-score $2 \times (0.375 \times 0.60) / (0.375 + 0.60) = 0.45 / 0.975 \approx 0.462$

What accuracy hid 93% sounds excellent - but the model misses 40% of all fraud, and only 37.5% of its fraud alerts are real.

The dummy comparison a model predicting 'not fraud' for everything scores 950/1000 = 95% accuracy - HIGHER than our real model, while catching zero fraud.

ANSWER **Accuracy 93% · Precision 37.5% · Recall 60% · F1 ≈ 0.462**

Which Metric? It Depends on the Cost of Being Wrong

There is no universally correct metric. There is a correct metric for YOUR cost structure.

Optimise RECALL when a miss is catastrophic

Cancer screening a False Negative sends a patient with a tumour home. A False Positive causes an unnecessary follow-up test. Missing is far worse.

Fraud detection an undetected fraud is direct financial loss; a false alarm costs an investigator's time.

Predictive maintenance a missed impending failure destroys the machine; a false alarm causes an unnecessary inspection.

Safety systems a pedestrian the car fails to detect is fatal; a phantom brake is uncomfortable.

How LOWER the decision threshold so more cases are flagged.

Optimise PRECISION when a false alarm is costly

Spam filtering a False Positive sends a genuine job offer to the spam folder, unseen. A False Negative just means one spam email in the inbox.

Content moderation wrongly removing legitimate speech has serious consequences.

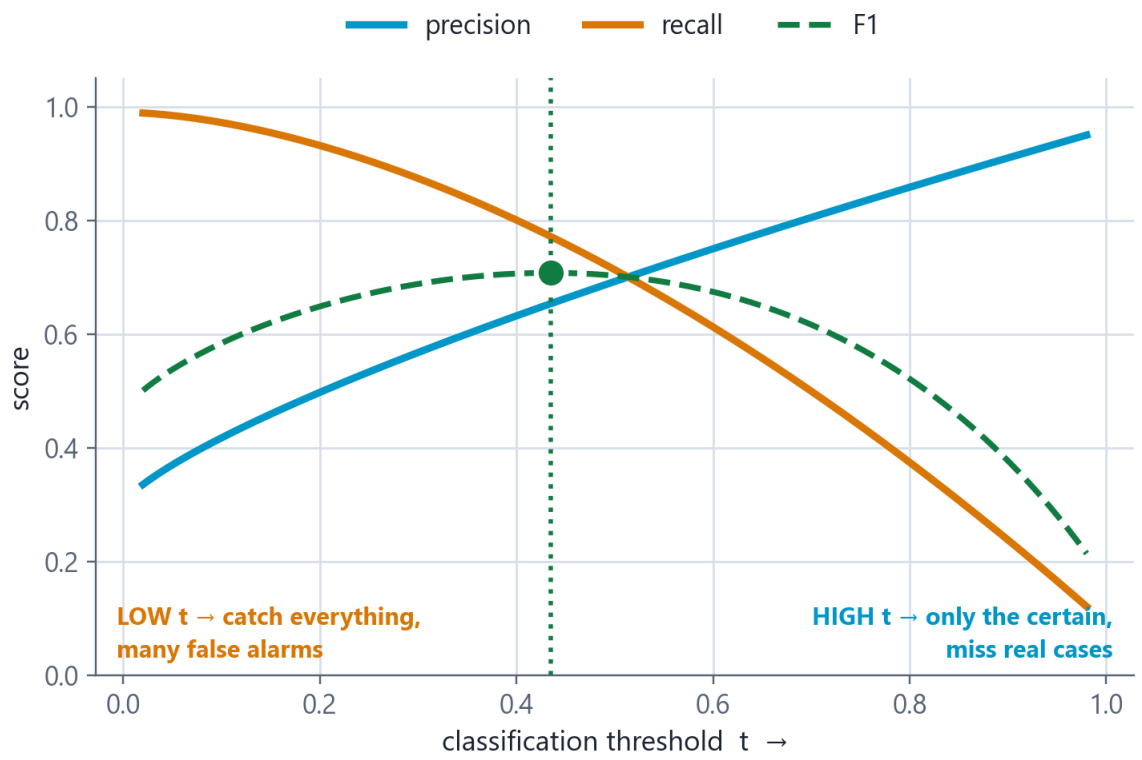
Targeted marketing contacting uninterested customers wastes budget and damages the brand.

Legal/criminal prediction a false accusation is far worse than a missed one - 'innocent until proven guilty' is a precision constraint.

How RAISE the decision threshold so only confident cases are flagged.

DECIDE A university builds a model to flag students at risk of failing, so tutors can offer support. Recall or precision - and what if the flag were used to deny scholarships instead?

Moving the Threshold Trades Precision Against Recall



HOW TO READ THIS FIGURE

Recall (gold) falls as the threshold rises - you catch fewer of the real cases.

Precision (blue) rises as the threshold rises - the ones you do flag are more often right.

They pull opposite ways you cannot maximise both by moving t. You can only choose where to sit.

F1 (green) their harmonic mean - one balanced compromise, peaked here.

The real decision which error costs more? That is a business question, not a technical one.

TAKEAWAY **The threshold is a dial you set from the cost of each error - 0.5 is a default, not an answer.**

ROBOTICS LINK

When Being Wrong Moves Something Heavy

UNIT 5

	predicted Pedestrian	predicted Clear
actual Pedestrian	**TRUE POSITIVE** **brakes** pedestrian seen, robot stops	**FALSE NEGATIVE** **collision** pedestrian missed - someone is hit
actual Clear	**FALSE POSITIVE** **phantom braking** nothing there - hard stop, rear-ended	**TRUE NEGATIVE** **proceeds** road clear, robot drives on

HOW TO READ THIS DIAGRAM

Same four cells as the fraud example. Completely different consequences.

False negative a person is struck. Unbounded cost.

False positive an emergency stop for nothing - and the vehicle behind does not expect it.

Both are injuries which is why you cannot simply push the threshold to one end.

This is why raw accuracy is close to meaningless here: 'no pedestrian' is right >99% of the time.

DID YOU KNOW?

Phantom braking - hard braking for an obstacle that is not there - is a documented, regulator-investigated failure mode in deployed driver-assistance systems.

THINK OF IT LIKE **a smoke alarm.** too sensitive and somebody removes the battery; not sensitive enough and it does not save anyone.

TAKEAWAY **On a robot, precision and recall are not scores. They are a choice about who gets hurt when the model is wrong.**

ROC Curve and AUC

Every other metric judges one threshold. ROC-AUC judges the model itself.

What the ROC curve shows

The axes True Positive Rate (Recall) on the y-axis against False Positive Rate, FP/(FP+TN), on the x-axis.

How it is drawn sweep the decision threshold from 1 down to 0 and plot the (FPR, TPR) pair at every threshold.

Why it is useful it evaluates the model across ALL thresholds at once, rather than committing to 0.5.

Shape to want a curve hugging the top-left corner - high recall with few false alarms.

The diagonal the line y = x is random guessing. Any useful model must sit above it.

What AUC means

Definition the Area Under the ROC Curve, between 0 and 1.

The intuition AUC is the probability that the model scores a randomly chosen positive higher than a randomly chosen negative.

AUC = 1.0 perfect separation of the two classes.

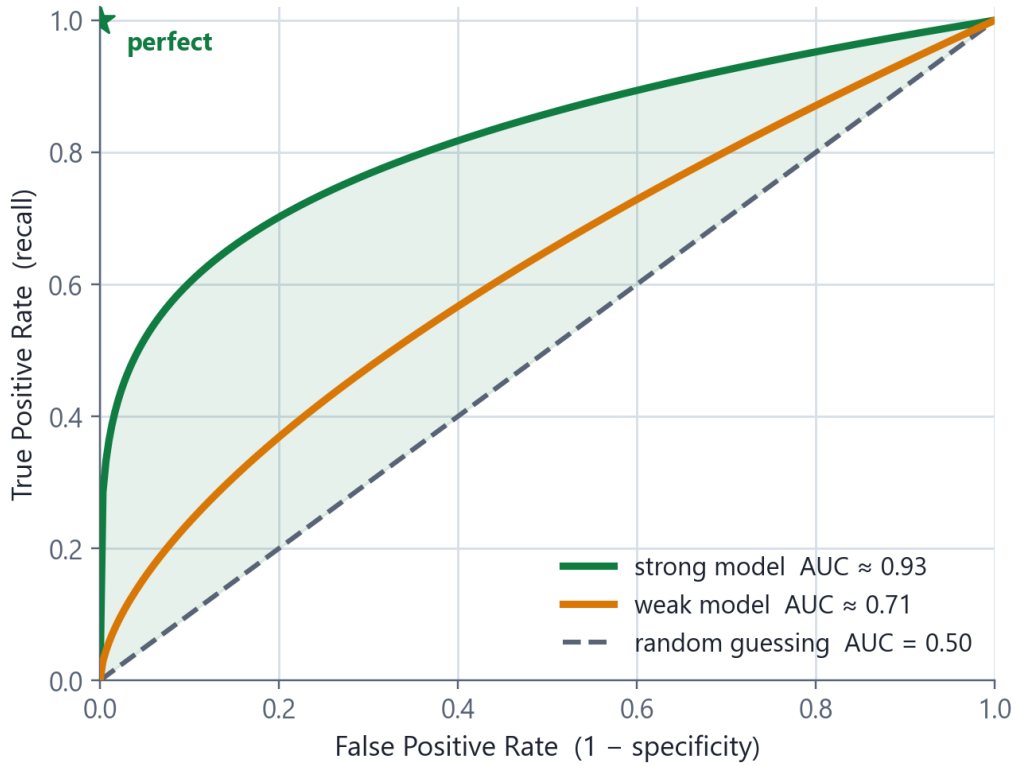
AUC = 0.5 no better than a coin flip.

AUC < 0.5 worse than random - the model has learned the classes backwards.

Caution on severely imbalanced data, ROC-AUC can look flattering. The Precision-Recall curve is more informative there.

ASK THE CLASS Two models both score 0.85 accuracy. Model A has AUC 0.92 and Model B has AUC 0.71. What does that tell you?

Reading an ROC Curve



HOW TO READ THIS FIGURE

Each curve traces one model across EVERY possible threshold - so it is threshold-independent.

Up and to the left is better. The star at (0, 1) is a perfect classifier.

The diagonal random guessing. AUC = 0.5. A model on this line has learned nothing.

AUC area under the curve - the probability the model ranks a random positive above a random negative.

Below the diagonal worse than random - which means your labels are probably inverted.

TAKEAWAY **AUC compares MODELS across all thresholds; the threshold plot chooses the operating point on one.**

ROBOTICS LINK

Module Metrics Multiply - Why the Stack Score Is Worse

UNIT 5



HOW TO READ THIS DIAGRAM

Each module tested alone, scores a respectable 95%.

Chained errors compound: $0.95 \times 0.95 \times 0.95 \times 0.95 \approx 0.81$.

Nineteen percent of the time, something in the chain is wrong.

Worse in practice a tracking error feeds a bad prediction - the failures are correlated, not independent.

So measure END TO END the number that matters is whether the whole stack behaved, not whether module three did.

DID YOU KNOW?

This is why autonomous-vehicle programmes report 'disengagements per mile' - a single end-to-end measure - rather than the accuracy of each perception module.

THINK OF IT LIKE **a relay team.** four fast runners and one dropped baton is still a lost race.

TAKEAWAY **Four modules at 95% do not give you a 95% robot. They give you roughly 81%.**

CHECK YOUR UNDERSTANDING

Quick Check - Section 2

UNIT 5

QUICK CHECK

A model for detecting a rare disease (2% prevalence) reports 97% accuracy, 0.90 precision and 0.15 recall. What is the correct assessment?

A. Excellent - 97% accuracy and 0.90 precision are both strong

B. It is nearly useless: it misses 85% of actual cases, and accuracy is inflated by the 98% healthy majority

C. The precision is too high, indicating overfitting

D. It needs more features to improve accuracy above 99%

ANSWER: B Recall 0.15 means the model detects only 15 of every 100 genuine cases - it misses 85%. For a disease screening tool that is a catastrophic failure. The 97% accuracy comes almost entirely from correctly identifying healthy people, who are 98% of the population. High precision here just means the few alerts it does raise are usually right - but it barely raises any.

03

SECTION

Model Improvement: Cross Validation and Hyperparameter Tuning

IN THIS SECTION

- Parameters vs hyperparameters - the distinction that matters
- Cross validation as the tuning engine
- Grid Search - exhaustive and slow
- Random Search - sampled and surprisingly better
- The train / validation / test discipline

Parameters vs Hyperparameters

Parameters are the answer. Hyperparameters are the question you asked.

Parameters - LEARNED from data

What they are the values the training algorithm discovers by minimising the cost function.

Examples the weights θ in linear and logistic regression; the split thresholds inside a decision tree; the centroids in K-Means.

Who sets them the algorithm. You never choose them by hand.

When during model.fit().

How many often thousands or millions.

Hyperparameters - CHOSEN by you

What they are configuration settings that control HOW learning happens.

Examples K in K-NN; max_depth in a tree; n_estimators in a forest; alpha (λ) in Ridge/Lasso; the learning rate α; K in K-Means; eps in DBSCAN.

Who sets them you - before training begins.

When when you construct the model object.

How to choose systematically, by cross-validated search. Never by guessing, and never by checking the test set.

SORT THESE θ_1 in linear regression · max_depth · the learning rate α · a tree's split threshold · n_neighbors · a K-Means centroid

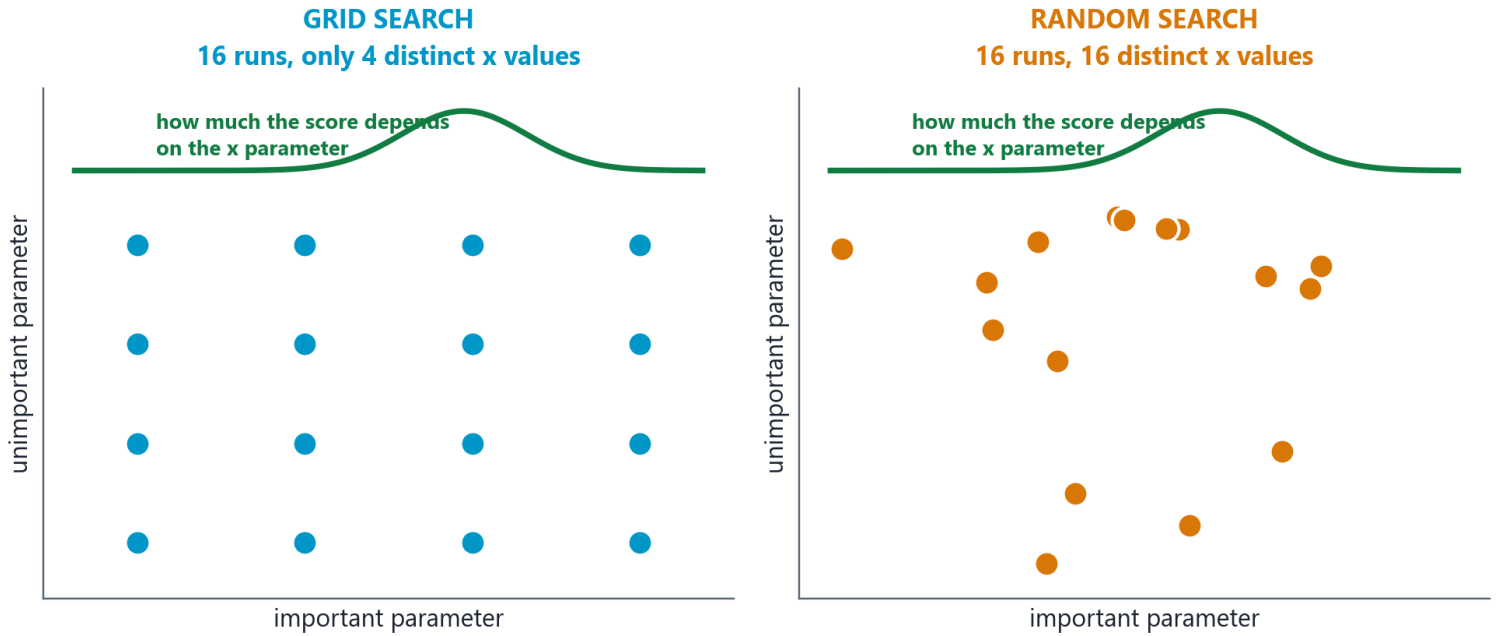
Grid Search vs Random Search

Both search the same space. One is thorough; the other is smart about its budget.

	Grid Search (GridSearchCV)	Random Search (RandomizedSearchCV)
How it works	Try EVERY combination in a predefined grid	Sample N random combinations from specified distributions
Cost	Product of all the options - grows explosively	Exactly N fits - you set the budget
Guarantee	Finds the best point ON THE GRID	No guarantee, but usually finds a near-best point
Continuous values	Must be discretised in advance, so you may miss the optimum	Samples continuously - can land anywhere in the range
Best when	Few hyperparameters, small ranges, plenty of compute	Many hyperparameters, or limited compute - the usual real case
The key insight	Wastes time exploring dimensions that do not matter	Explores more distinct values of the parameters that DO matter

CALCULATE n_estimators has 4 options, max_depth has 5, min_samples_split has 3, and you use 5-fold CV. How many model fits does Grid Search perform?

Why Random Search Usually Beats Grid Search



HOW TO READ THIS FIGURE

Both panels sixteen runs. Identical compute budget.

Grid search 16 runs, but only FOUR distinct values of each parameter.

Random search 16 runs, and SIXTEEN distinct values of each parameter.

The green curve how much the score depends on the horizontal parameter. The vertical one barely matters.

The consequence grid search wasted 12 runs re-testing four x values; random search sampled the axis that mattered.

TAKEAWAY **Same budget. Random search explores the parameter that matters four times more finely.**

The Train / Validation / Test Discipline

COMMON PITFALL · Three sets, three distinct jobs - do not mix them

TRAINING set (~60-70%) the model learns its PARAMETERS here. Seen many times.

VALIDATION set (~15-20%) you choose HYPERPARAMETERS here, and compare models. Seen many times, but never learned from directly.

TEST set (~15-20%) the final, honest estimate of performance. Seen EXACTLY ONCE, at the very end.

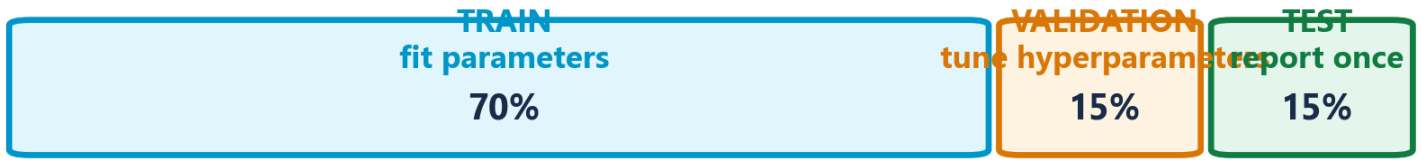
Where cross-validation fits K-fold CV on the training data replaces a separate validation set - it gives you K validation folds for free. The test set stays untouched either way.

The failure mode checking test accuracy, tweaking a hyperparameter, and repeating. You have now optimised against the test set through your own decisions - the number is no longer honest.

The name for it this is why competition leaderboards have a private split that nobody sees until the contest closes.

ASK THE CLASS You run GridSearchCV, then check test accuracy, then widen the grid and run again, then check test accuracy again. What have you just done?

Three Sets, Three Jobs



tune on validation as often as you like - open test exactly once

HOW TO READ THIS FIGURE

Train the model fits its parameters here.

Validation you choose hyperparameters here. Use it as often as you like.

Test opened once, at the very end, to report a number you can defend.

The failure tuning against test makes your reported score optimistic - and you cannot un-see it.

With cross-validation the train and validation blocks become the k folds; test still stays sealed.

TAKEAWAY **If you tuned on it, you cannot report on it.**

04

SECTION

Introduction to Explainable AI

IN THIS SECTION

- Why black-box models are a business and legal problem
- Interpretability vs accuracy - the real tradeoff
- LIME - local surrogate explanations
- SHAP - Shapley values from game theory
- Global vs local explanations

Why Explainability Matters

KEY CONCEPT · **A model you cannot explain is a model you cannot defend**

The legal driver GDPR and similar regulations constrain solely automated decisions with legal or significant effect, and support a right to meaningful information about the logic involved. 'The neural network decided' is not an adequate answer.

The trust driver clinicians will not act on a diagnosis they cannot interrogate. Adoption fails without explanation, however accurate the model.

The debugging driver explanations reveal shortcut learning. If a pneumonia model's explanation highlights the scanner watermark rather than the lung, you have found a bug you could not otherwise see.

The fairness driver you cannot audit for bias without knowing which features drive decisions. This is how Unit 1's fairness pillar becomes actionable.

The improvement driver understanding why a model fails tells you which features to engineer next.

The tradeoff the most accurate models - deep networks, large ensembles - are the least interpretable. That tension is the whole field.

ASK THE CLASS Would you accept a 2% drop in accuracy to make a model fully interpretable? Does your answer change for a film recommender versus a medical diagnosis?

LIME and SHAP - The Two Standard Techniques

Both answer 'why THIS prediction?' - LIME by local approximation, SHAP by fair credit allocation.

LIME - Local Interpretable Model-agnostic Explanations

The idea to explain ONE prediction, fit a simple interpretable model (usually linear) that approximates the black box in the small neighbourhood around that input.

How perturb the input many times, record what the black box predicts for each variant, then fit a weighted linear model to those results.

Output 'this loan was rejected mainly because debt-to-income was high and credit history was short.'

Strengths fast, model-agnostic, works on text and images as well as tables.

Weaknesses explanations can be unstable - re-run it and the explanation may shift. It is LOCAL only.

SHAP - SHapley Additive exPlanations

The idea borrowed from cooperative game theory. Treat each feature as a player in a game whose payoff is the prediction, and compute each player's fair share of the credit.

The Shapley value φ_i the average marginal contribution of feature i across all possible orderings of the features.

Guarantees the contributions sum exactly to the prediction, and the method is consistent - unlike LIME.

Output force plots for one prediction; summary plots for the whole dataset. Both LOCAL and GLOBAL.

Weakness computationally expensive in general - though TreeSHAP is fast for tree ensembles.

Case Study: Credit Risk and the Adverse Action Notice

REAL-WORLD APPLICATION · **Explainability as a legal deliverable, not a nice-to-have**

The situation a bank uses a gradient boosting model to decide mortgage applications. It is accurate, and it is a black box.

The obligation when an application is declined, the applicant is entitled to know the principal reasons. In several jurisdictions this notice is legally mandated.

The solution run SHAP on the individual declined application. The force plot ranks each feature's contribution to that specific decision.

The output 'Debt-to-income ratio contributed +42% toward decline; 5 recent credit inquiries contributed +18%; length of employment contributed −7% toward approval.'

Why SHAP rather than global importance global feature importance tells you what the model relies on ON AVERAGE. The applicant needs to know about THEIR application - that requires a local explanation.

The bonus the same explanations let the bank audit for bias, checking whether protected attributes or their proxies are driving decisions.

UNIT V CONSOLIDATION

Activity: Choose the Metric, Justify the Threshold

UNIT 5

PAIR WORK · 10 minutes

Four deployed systems. For each: name the metric you would optimise, say whether you would raise or lower the decision threshold, and state which error is more costly.

1

System A

- Airport security screening for prohibited items
- Millions of bags per year
- Human inspectors handle every alert

2

System B

- An email client's spam filter
- Users rarely check the spam folder
- Business correspondence flows through it

3

System C

- Predicting equipment failure in a power plant
- Unplanned shutdown costs ₹2 crore per event
- An inspection costs ₹50,000

4

System D

- Recommending which of 10,000 films a user might enjoy
- The user sees 20 recommendations
- No safety or financial consequence

DELIVERABLE One metric per system, plus one sentence naming the costly error and the threshold direction. Be ready to be challenged.

COMMON MISTAKES

Common Mistakes to Avoid

Every one of these is a real error students make in this unit - and every one is avoidable.

MISTAKE 1

Quoting accuracy on imbalanced data

- A do-nothing model can score 99%
- Report precision and recall instead

MISTAKE 2

Treating 0.5 as the threshold

- It is a default, not a law
- Set it from the cost of each error

MISTAKE 3

Tuning against the test set

- Your reported score becomes the max of many draws
- If you tuned on it, you cannot report on it

MISTAKE 4

Quoting best_score_ as the result

- That is the cross-validated TRAINING score
- The test score is the honest one

THE FOUR THAT COST MARKS

✗ accuracy on a rare event

✗ 0.5 because it is the default

✗ tune on test

✓ if you tuned on it, do not report it

Research Lens

Where this topic is heading - so you know what you are being prepared for.

DIRECTION 1

Calibration, not just ranking

- Does '0.8' actually mean 80% of the time?
- Matters enormously in medicine and risk
- AUC says nothing about it

DIRECTION 2

Evaluating generative models

- No confusion matrix for free-form text
- LLM-as-judge, human preference, task benchmarks
- Actively contested

DIRECTION 3

Fairness metrics that conflict

- Several reasonable definitions of 'fair'
- It is provably impossible to satisfy all of them
- So the choice is a value judgement

WHERE THIS IS GOING

→ does 0.8 really mean 80%?

→ how do you score free text?

! fairness definitions provably conflict

UNIT V RECAP

Unit V - The Six Ideas Worth Remembering

UNIT 5

1. **Regression metrics differ in how they punish large errors**
   MAE is robust and linear; MSE/RMSE punish big errors quadratically; R^2 gives the proportion of variance explained. RMSE ≥ MAE always.

2. **Plain R^2 can be gamed by adding features**
   It never decreases when a predictor is added, so it always favours complexity. Adjusted R^2 penalises p - use it to compare models with different feature counts.

3. **The confusion matrix is the foundation of everything**
   TP, TN, FP (Type I), FN (Type II). Read the labels as: was the model right, and what did it predict? All four classification metrics come from these cells.

4. **Accuracy lies on imbalanced data**
   A dummy model predicting the majority class can beat a real model. Precision = TP/(TP+FP) asks 'do I trust the alert?'; Recall = TP/(TP+FN) asks 'what did I miss?'

5. **The right metric follows the COST of each error**
   Optimise recall when a miss is catastrophic; precision when a false alarm is. ROC-AUC judges the model across all thresholds, not just 0.5.

6. **Tune hyperparameters with CV - never on the test set**
   Parameters are learned, hyperparameters are chosen. Grid Search is exhaustive; Random Search often wins at equal budget. And explain the result with LIME or SHAP.

NEXT UP Unit VI - AI for Intelligent Systems and Robotics: computer vision, NLP and conversational AI, reinforcement learning, and autonomous robots.